Suchit Srivastava

ai23btech11027@iith.ac.in | suchit2309@gmail.com | +91 9873450819

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.99
Lotus Valley International School Gurgaon	Class XII (CBSE)	2022	95.20%
Lotus Valley International School Gurgaon	Class X (CBSE)	2022	96.80%

SCHOLASTIC ACHIEVEMENTS

- First place in Waveform Wars (Elan & Envision'25) conducted in partnership with Australian government
- Achieved 6th place in Inter IIT Tech Meet 13.0, at IIT Bombay for the IGDC problem statement.

WORK EXPERIENCE

Policy Bazaar Internship, 2025 Summer:

- Developed a modular workflow to improve doctor patient interactions using ASR and other sequential models
- Tested the workflow with many publically available models like whispr and NeMo ASR
- Provided results for transcription, translation, transliteration models to find the optimal set of models

Quadeye Internship, 2026 Summer:

- Created a scalable agentic workflow to analyse codebases to help new engineers onboard onto existing project
- Mapped out multiple workflows involving security constraints through agent sandboxing and estimating user proficiency with foundational math.
- Enhanced workflows with interpretability, failsafes, appropriate resumes and proper modularity.
- Also worked on improving anomaly detection in huge streams of data using models like ARIMA

PROJECTS

RISC-V Simulator

- Implemented an assembler: converting assembly code to 32 bit instruction binary
- Implemented a simulator with 32 registers and limited memory which takes instruction binary to simulate tasks
- Added customizable cache configurations with LRU, FIFO and Random replacement policies along with support for direct-mapped, fully-associative, and set-associative mapping

Tetris Playing Bot

- Simulated Tetris in Python using matplotlib and numpy
- Found heuristics to evaluate Tetris board positions to analyse which positions are more suitable at each step and what factors contribute to a good game of Tetris
- Applied Genetic Search Algorithm (GSA) to properly weigh the found heuristics and find the optimal next step using this weighting.

Noise cancellation in Audio Files

- Evaluated audio spectrograms in MATLAB to get analyse waveforms of audio samples
- Found methods to find regions of pure noise using Short Time Energy (STE) and decibel values.
- Filtered out noise using the frequencies from found noise set.
- First place in Waveform Wars Round 1 : Signal Processing competition at "Elan & nVision" (IITH's flagship fest). Conducted in partnership with Australian government.

Rooftop Detection and Annotation

- Created a program to detect rooftops in an image and annotate a polygon around them with a provided dataset

- First place in Waveform Wars Round 2 conducted in partnership with Australian government.

Inter IIT Tech Meet 13.0 - IIT Bombay

- Created a video game (Dark Voyager) with unique visual mechanics to navigate environment taking inspiration from LiDAR and echolocation.
- Came up with the idea, the plan and created the game with a team of 4 people in 2 weeks
- Achieved 6th place overall in the problem statement

Stress Granule Detection | Computer Vision & Bioimage Analysis

- Developed an automated pipeline for **stress granule detection and quantification** in fluorescence microscopy images using InstanSeg and DBSCAN clustering.
- Integrated **3D cell segmentation and granule-level analysis** to characterize spatial distributions and morphological properties of stress granules.
- Implemented image processing and tracking workflows using **Python, NumPy, and scientific computing libraries** for cellular imaging analysis.

Perceptual Straightening in Video Diffusion:

- Developed an inference-time method to improve **temporal consistency** in AnimateDiff-generated videos using perceptual features and trajectory smoothing.
- Implemented **adaptive perceptual smoothing, kernel regression, and ensemble fusion** to reduce jitter and temporal artifacts.
- Optimized the diffusion pipeline for **RTX 4060 (8GB)** using mixed precision, CPU offloading, and VAE tiling.

Comparison of MAB Algorithms:

- Implemented and compared **10+ Multi-Armed Bandit algorithms**, including Epsilon-Greedy, UCB variants, MOSS, KL-UCB, Bayes-UCB, and Thompson Sampling.
- Built a **modular bandit simulator** with configurable reward distributions, agents, and experiment settings to study exploration-exploitation behavior.
- Evaluated algorithms using **reward, regret, arm-selection, and convergence analysis** across simulated Bernoulli bandit environments.

NLP-Driven Financial Decision Modeling:

- Engineered a modular evaluation pipeline integrating FinBERT, FLANG, and LSTM models to extract financial signals and forecast asset performance for equities and crypto.
- Developed an end-to-end application to simulate automated trading outcomes, map model confidence, and benchmark algorithmic strategies.

POSITIONS OF RESPONSIBILITY

Overall Head at Glitch (The Video Game Club of IITH) 2025-26:

- Ideated and spearheaded the first inter-hostel browser based competitive game with live leaderboards and results.
- Led a team of game developers to release the first multiplayer game of the club.
- Managed a team of Esports enthusiasts to conduct many gaming tournaments on campus throughout the year .

Core at Kludge - Information and Cybersecurity Club of IITH (2024-25)

Core at Glitch- Game Development Club of IITH (2024-25)

EXTRACURRICULAR

Coordinated AI-Day Musical Performances (2025)

Member of Vibes - Music Club of IITH (2023-Ongoing)

Founded member of the first robotics team in my school LVIS in the year 2021, participated in an international robotics challenge **FIRST TECH CHALLENGE** in 2021.

SKILLS

- **Languages:** Python, C, SQL, JavaScript
- **Libraries and Frameworks:** Numpy, Pandas, Pytorch, Matplotlib, ReAct, Django

- **Other Tools and Softwares:** Unity (Game Development Engine), LaTeX, Git, ROS2, MuJoCo, Gazebo

RELEVANT COURSES

Foundational Mathematics: Discrete Mathematics, Probability, Matrix Theory, Metric Spaces, Linear Systems and Signal Processing, Convex Optimisation, Information Theory, Applied Statistics, Calculus I, Calculus II, Series of Functions.

Artificial Intelligence: Artificial Intelligence, Natural Language Processing, Computer Vision, Reinforcement Learning, Online Learning, Programming for AI, Foundations of Machine Learning

Computer Science: Computer Architecture, Database Management Systems I, Database Management Systems II, Algorithms, Data Structures and Applications, Operating Systems I, Compilers I, Fraud Analytics